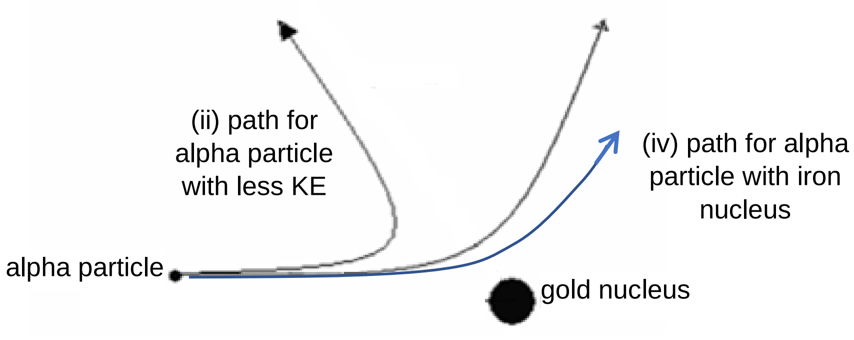
5ai	Gold nucleus and alpha particle are positive charged. Thus, the gold nucleus repels the alpha particle.	A1
5aii	Greater deflection of final path and final path has smaller distance of closest approach 	A1

5aiii	A thin gold foil is required as to ensure one alpha particle to collide with a single gold nucleus or to avoid multiple collisions of one alpha particle with many gold nuclei.	A1
5aiv	Lesser deflection of final path and final path has greater distance of closest approach	A1
5bi	As alpha particle approaches nucleus, KE is converted to EPE, U. Hence alpha particle must possess minimum energy *U* to be this close to Li By conservation of energy, Loss in KE = Gain in EPE $U = (1/4\pi\varepsilon_0)\,(Q_1 Q_2 / r)$ $= (9\times10^{9}) \times [(2\text{x}1.6\times10^{-19}) \times (3\text{x}1.6\times10^{-19}) / (4.2\times10^{-15})$ $= 3.3 \times 10^{-13}$ J	 M1 M1 A0
5bii	Maximum possible energy of a neutron would occur when Boron does not have kinetic energy. Maximum Energy of neutron = (Rest mass energy of reactants)+(K.E. of alpha) – (Rest mass energy of products) $= (4.0015+7.0144)uc^2 + 3.3 \times 10^{-13} - (10.0011+1.0087)uc^2$ $= 1.24 \times 10^{-12}$ J	M1 M1 A1
5ci	Photoelectron is an electron emitted from a metal when light of certain frequency (or frequency higher than threshold frequency of metal / sufficient energy) is shone on the metal. β-particle is a fast-moving electron emitted from an unstable nucleus in radioactive decay.	B1 B1
5cii	$dN/dt = -\lambda N$	A1
5ciii	unit of $\lambda = s^{-1}$	A1
5civ	$A = A_o e^{-\frac{ln2}{t_{1/2}}t}$ $2.8\,x\,10^3 = A_o e^{-\frac{ln2}{4.2x10^4}(7.2x10^4)}$ $A_o = 9.19 \times 10^3\ Bq$	 M1 A1

Section B

6ai	It is rate of change of velocity	A1